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Multi-Task Feature Decoupling Network with clear division of labor for vehicle component detection

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ABSTRACT

To enhance the precision and stability of vehicle component detection for intelligent claims assessment and intelligent vehicle detection, it is necessary to address the problem of missing detection and false detection in complex scenes caused by feature similarity between classes and insufficient feature extraction. In this paper, we propose a decoupling network architecture, named the Multi-Task Feature Decoupling Network (MTFDN), which is based on a two-stage multitasking algorithm. Three improved modules have been designed and integrated into the architecture for this purpose. First, we design an improved Parallel Dynamic Attention Module (PDAM) to facilitate a more scientific and logical acquisition of features and localization of regions of interest. Then, we introduce an improved Bidirectional Fusion Feature Pyramid Network (BF-FPN) module in the segmentation task, followed by the design of a Bidirectional Hybrid Dilated Fusion Feature Pyramid Network (BHDF-FPN) module in the detection task. The BF-FPN and BHDF-FPN modules are used to meet the different requirements of different tasks for features after decoupling. The experimental results demonstrate that, in comparison with the baseline model, MTFDN exhibits superior performance in terms of detection and segmentation on two vehicle-related datasets, namely the Vehicle Scene Dataset (VSD) and the Vehicle Component Dataset (VCD). Moreover, MTFDN demonstrates robust generalization capabilities when integrated with other two-stage algorithms, significantly enhancing the detection and segmentation performance of the algorithm itself, thereby highlighting its superiority in this domain.

1. Introduction

The rapid growth of the automobile industry has not only enhanced people's quality of life but also bolstered the development of the social economy [1]. However, the associated urban traffic issues, such as traffic congestion and driverless driving safety risks, have become increasingly severe, demanding resolution [2–4].

In traditional automobile insurance practices, vehicle scene investigation constitutes the most critical aspect of the claim process. However, this procedure is generally undertaken by insurance company staff, necessitating significant investment in human and material resources [5]. Following an accident, several steps, including alarm, liability identification, on-site investigation, and photography by insurance company personnel, must be completed before the damaged vehicle can be moved, exacerbating the already extensive urban traffic pressure. Therefore, to alleviate traffic congestion and streamline the automobile insurance compensation process, an intelligent claims

assessment system is necessary to analyze images of damaged vehicles and determine the damaged components, types of damage, and their severity. The identification of damaged components is completed through component recognition, and accurate positioning of the components is the basis and guarantee for precise damage assessment. Insurance companies need to ascertain which part of the vehicle has been damaged, in order to facilitate the subsequent replacement of the damaged parts. This approach will reduce the loss assessment cost, simplify the entire industry process, and drive technical progress in the auto insurance domain [6].

In the realm of driverless technology research and development, vehicles are critical identification targets and the primary threatening obstacles in driverless driving [4,7,8]. Detection based on global vehicle information leads to false alarms or missed detection due to factors such as occlusion, illumination, and detection targets not having global information in sample training, which easily causes vehicle collisions and other traffic accidents, posing a very large threat to the safe

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operation of unmanned vehicles [9]. Vehicle detection methods based on components involve partitioning the entire vehicle into several constituent parts and transforming the vehicle detection task into a vehicle component detection task. Consequently, in scenarios where there is partial vehicle occlusion, it is possible to determine whether a vehicle is present by evaluating the confidence scores of other vehicle components, such as headlights, or front and rear wheels Bileschi and Heisele [10], Leung [11] and Ruan et al. [12]. By leveraging local component information, component-based vehicle detection methods can mitigate the dependence on model generality and instead rely on component detection accuracy. This approach is particularly suitable for complex road traffic scenarios [13].

In summary, if the missing and false detection problems in complex scenarios can be addressed to improve vehicle component detection accuracy, it can be integrated into intelligent claims assessment, vehicle detection, and other applications as a modular component. Therefore, the development of methods to enhance the performance of vehicle component detection and segmentation is of great practical significance and broad application potential. The advancement of this technology has a crucial role in reducing labour costs and facilitating urban transportation.

However, there are often internal and external factors affecting the vehicle component detection accuracy:

- Internal factors include very similar textures, colors, and so on among vehicle component categories.
- External factors include dark or reflective light in shooting scenes, multivehicle scenes, partially blocked components, blurred picture quality, etc.

To address the prevalent missing and false detection issues in complex vehicle component detection scenes in the vehicle industry, we explored a common optimization direction for a top-down two-stage multi-task network to address these issues at the feature level. In contrast to prior work, our approach focuses on the unique feature requirements of each task and does not rely on sharing the same feature map across multiple tasks. Specifically, we aim to decouple task features and implement targeted designs in the feature extraction module.

The main contributions of this paper can be summarized as follows:

1. We propose a Multi-Task Feature Decoupling Network (MTFDN), which extends the idea of decoupling classification and regression to task decoupling, transforming task constraints into favorable drivers. This decoupling method can be extended to other two-stage multi-task algorithms and get great performance improvement, demonstrating exceptional capabilities in generalization and integration.
2. Two improved feature pyramid network modules are introduced into the MTFDN architecture to cater to the feature requirements of both the detection and segmentation tasks. A Bidirectional Hybrid Dilated Fusion Feature Pyramid Network (BHDF-FPN) is tailored to serve the detection task, while a Bidirectional Fusion Feature Pyramid Network (BF-FPN) focuses on the segmentation task, thereby making the feature extraction more task specific.
3. To make feature learning more scientific and interpretable, we propose a Parallel Dynamic Attention Module (PDAM), which assigns different levels of importance to channel and spatial features during different training stages, allowing the model's learning process to have human-like sequentiality. This mechanism facilitates a more scientific and logical acquisition of features and localization of regions of interest.
4. In the context of vehicle component detection, MTFDN effectively addresses challenges posed by inter-class feature similarity, inadequate feature extraction, and varying demands within the detection-segmentation tasks. Additionally, its applicability can extend to a broader spectrum of task decoupling scenarios.

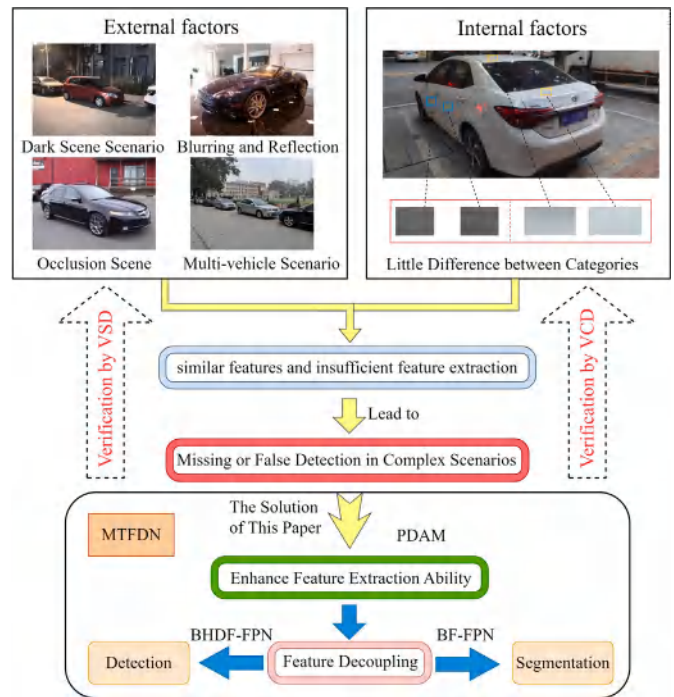


Fig. 1. The vehicle component detection problem and the proposed solution.

In this study, we establish two kinds of datasets for internal and external factors: VSD specifies 5 component categories with significant differences among them. Most of the pictures in VSD are long-distance pictures of vehicles, which are used to reflect the improvement of external factors in the comparison experiment. VCD specifies 59 component categories, among which most pictures are close-range pictures of vehicles, which are used to reflect the improvement of external factors in the comparison experiment. Fig. 1 shows the problem of vehicle component inspection and the solution proposed in this article.

The rest of this paper is organized as follows. The second section introduces the relevant research on vehicle component detection and the past application of the decoupling idea, while the third section provides a comprehensive and detailed description of the overall framework and each module. The fourth section explains the experimental setup, evaluation criteria, dataset preparation, experimental validation process, and application extension. Finally, the fifth section summarizes the findings and proposes future work.

2. Related work

2.1. Vehicle component detection

Vehicle component detection method based on prior knowledge [14–17]. The methods involve building a spatial geometric correlation model based on prior knowledge, screening potential components from vehicle images, extracting matching features from the selected regions, and training a classifier to achieve component recognition. This approach effectively avoids the extensive and repetitive operations of traditional manual feature methods. However, it still requires manual characterization, which may lead to unsatisfactory feature descriptions when there is more than one component.

Vehicle component detection method based on a convolutional neural network (CNN). At present, the object detection methods based on convolutional neural network (CNN [18]) can be divided into two categories: methods based on region proposals [19–22] (two-stage detector) and regression-based methods [23,24] (one-stage detector). In two-stage methods, a region proposal network (RPN [19]) is used

in the first stage to generate region proposals on the feature maps extracted by the CNN. In the second stage, these regions are classified and regressed to the boundary boxes. Representative two-stage methods include Faster R-CNN [19], R-FCN [20], and Couple-Net [21]. In one-stage methods, the two stages are integrated into one, and the RPN stage is omitted. Categorization and bounding box regression are carried out simultaneously in the detector. Representative methods include SSD [22], YOLO [23] and RetinaNet [24].

When the object detection algorithm is applied to vehicle component detection, the algorithm universality may fall short of addressing the data particularity, necessitating targeted design. Wu [25] utilized the K-means++ algorithm to reanchor the YOLOv3 [23] network for vehicle component detection. However, the network's ability to address interclass similarity issues in component detection was not optimized, leading to relatively high misjudgment rates between component categories. Hence, in light of the challenges associated with missing and false detections in complex scenes arising from feature similarity among vehicle components and insufficient feature extraction, researchers often utilize vehicle component positional relationships, component cooccurrence statistics, and orientation information to constrain and evaluate test results, thereby mitigating the issue at hand. For instance, Juan [26] carried out vehicle component feature point detection using a three-level cascade convolutional neural network and used vehicle component relative position information and key point geometric relations as constraints. However, the multi-cascade network proved too complex, with a high number of parameters, poor generalization, and embeddability. On the other hand, Jianxiang [27] explicitly incorporated the spatial position relationship between components into CNN and regional proposal network structures using an angular distance network. Moreover, vehicle component spatial position statistical information was taken as prior knowledge to constrain the classification results. However, the network length was extended, resulting in a low detection rate. In addition, feature extraction at the lower level was not optimized, leading to poor detection results for small components, which fails to meet application requirements.

In light of the aforementioned issue of similar component characteristics, the commonly used approach of utilizing prior information for post-processing often suffers from poor generalization capability across different datasets. Thus, it is imperative to address the problem fundamentally from the perspective of task features. Moreover, when there are numerous categories of components to be detected, if only the bounding box is used to mark the detected vehicle components, the prediction box will be too dense and its edges will not describe the component profile well, which leads to the application of parts detection in intelligent claims assessment and other fields, far from meeting the industry demand. Therefore, we need to introduce the pixel-level classification method into the components detection.

2.2. Instance segmentation

In contrast to traditional image segmentation methods, instance segmentation technology not only breaks through plain and intuitive visual meaning but also overcomes the limitation of conveying higher-level semantic information that cannot be combined. Instance segmentation can distinguish different instances based on image pixel-level classification, which can also be divided into one-stage algorithms, Query-based algorithms, and two-stage algorithms.

The one-stage algorithms [28–31] do not rely on the traditional region proposal and directly generate the instance segmentation results of pixels. One is inspired by anchor-based detection models such as YOLO [23], including SOLO [28] and YOLACT [29]. For instance, SOLO [28] assigns categories to each pixel within an instance based on attributes such as spatial location and size. Similarly, YOLACT [29] employs a linear combination of masks and predicted coefficients for each instance, followed by trimming with the predicted bounding

box. The other is inspired by anchor-free detection models such as FCOS [32], including PolarMask [31].

The query-based instance segmentation methods (SOTR [33], SOLQ [34], Mask2Former [35], FastInst [36]) are inspired by DETR and do not rely on convolutions. Instead, they utilize versatile and powerful attention mechanisms combined with a set of learnable queries to infer object classes and segmentation masks, aiming to unify detection and segmentation tasks. For instance, Mask2Former [35] simplifies the workflow of instance segmentation by adding a pixel decoder and a Masked Attention Transformer decoder to the backbone. However, there still exist significant differences between detection and segmentation models, which hinder task and data collaboration between detection and segmentation. As a result, the state-of-the-art query-based instance segmentation model, Mask2Former, still lags behind classical models based on HTC++ [37] and Swin-V2-G [38].

Meanwhile, the two-stage algorithms can be subdivided into bottom-up and top-down methodologies.

Within the bottom-up algorithm, initial pixel-level semantic segmentation is performed, succeeded by the differentiation of distinct instances through techniques such as clustering and metric learning, as demonstrated by SGN [39] and SSAP [40]. The algorithms in this category learn embeddings and instance-aware features for each pixel and then use post-processing techniques to group them into instances based on their embedding characteristics. Consequently, these methods effectively preserve positional and local coherence information. However, the main drawback of bottom-up models is the instability of clustering, which leads to poor generalization capability across different datasets with varying scenes.

On the other hand, the top-down algorithm commences with instance detection, followed by subsequent semantic segmentation of the identified instances. Illustrative implementations of this approach include FCIS [41], Mask RCNN [42], PANet [43], Mask Scoring RCNN [44], Cascade Mask R-CNN [45], Hybrid Task Cascade (HTC [37]) and Mask Transfiner [46]. Typically, the top-down algorithmic strategy exhibits superior efficacy compared to its bottom-up counterpart. This paradigm consists of a detector for identifying and locating all objects and a mask branch for generating segmentation masks. The success of this segmentation approach is attributed to translation equivariance and location. However, it also faces the following challenges: (1) Due to limited receptive fields, CNNs lack the coherence of features in high-level visual semantic information to associate instances, resulting in suboptimal segmentation results for dense objects. (2) The segmentation quality relies on the object detector and performs poorly in complex scenes.

Regarding (1), enlarging the receptive field to obtain semantic information inevitably sacrifices detailed information. Regarding (2), improving segmentation quality requires focusing on detailed information, while insufficient semantic information makes it difficult to obtain correct class information.

In the era of large models, Transformers [33] have the ability to easily capture global features and naturally model long-range semantic dependencies. Therefore, Transformer-based models [33,34,36,38,46,47] can better distinguish overlapping instances with the same semantic category, making them more suitable for visual segmentation tasks compared to CNNs. From these perspectives, Transformers seem to be able to address challenges (1) and (2) effectively.

However, these Transformer-based methods still have limitations [47]. On the one hand, Transformers perform poorly in extracting low-level features, leading to incorrect predictions for small objects. On the other hand, due to the extensive feature mapping, they require a large amount of memory and time, especially during the training phase. Additionally, when trained on insufficient data (ImageNet as a medium-sized dataset), Transformer models perform worse than ResNet models with similar parameter sizes. This is because Transformers struggle to learn some of the inherent inductive biases present in CNNs, such as translation equivariance and locality [33]. Moreover, due to the large

number of vehicle component categories, annotating the parts of the entire vehicle is extremely complex and tedious. It is challenging to complete large-scale vehicle component data sets, and Transformer, as a feature extraction network, is not the best choice to achieve better results with small-volume vehicle component data sets.

Thus far, challenges (1) and (2) reflect the dilemma between capturing details and overall understanding in features, indicating an irreconcilable contradiction in the feature requirements for detection and segmentation tasks. For the top-down instance segmentation learning paradigm, the introduction of decoupling ideas seems to be a promising solution.

2.3. Decoupling in object detection

When the idea of decoupling was initially applied to detection, it was used for decoupling classification and bounding box regression tasks. Jiang et al. [48], Wu et al. [49] and Song et al. [50] first proposed this problem: The classification and bounding box regression tasks carried out by a shared header detection network at the same time often have a mutual influence problem between the two task types. The visual features generating better classification scores cannot generate bounding boxes with higher accuracy at the same time, and the classification confidence of candidate boxes is biased from the location confidence. Therefore, they use decoupling to construct new detection networks that generate location confidence and learn the prediction candidate box, obtain the regression box with a more accurate location, and process the location regression problem as an optimization task to achieve a more accurate detection effect. However, the problem of task coupling appears not only in classification and regression tasks but also in detection and segmentation tasks. Therefore, the application of the decoupling idea can be further expanded here.

3. Network architecture and module design

3.1. Network architecture design

In the context of segmentation tasks, achieving high-precision target detection is a crucial prerequisite for successfully completing the task. Unlike detection tasks, which need to integrate high-level semantic information and low-level detail information simultaneously, segmentation tasks predominantly rely on detailed information with supplementary input from semantic information. As a result, these two tasks exhibit varying sensitivities towards features at different levels.

We have noticed that most segmentation detection models share a feature extraction network for both detection and segmentation tasks. However, this shared feature often fails to meet the different feature requirements of each task. Inspired by the decoupled head detection network in the literature [48–50], we further extend the application of the decoupling concept, expanding the decoupling of classification and regression to task decoupling. In order to fully utilize the distinct characteristics of detection and segmentation, this paper proposes an improved decoupled network architecture – the Multi-Task Feature Decoupling Network (MTFDN). Two feature extraction fusion networks that meet the different task requirements work in parallel while mutually promoting each other. This effectively transforms the problem that often constrains segmentation performance in two-stage algorithms (high-precision object detection is a crucial prerequisite for successful segmentation) into a beneficial driving relationship. This transformation is ingenious: when the detection branch does not need to consider the feature requirements of the segmentation task, it can focus on the size and location of the targets in the case of greatly expanding the receptive field, avoiding missed detections and false alarms. Meanwhile, the segmentation branch balances detailed information and semantic information for boundary delineation and mask allocation. At this point, more accurate detection outputs (target positioning and anchor boxes) will ensure the accurate output of segmentation masks.

MTFDN, as a top-down two-stage multi-task algorithm framework, includes three modified modules. As a two-stage algorithm, Mask RCNN [42] serves as the baseline model in this study, but it is not the only algorithm applicable to this research. This paper presents an attempt to decouple the feature extraction networks for distinct tasks, introducing the MTFDN architecture. Specifically, we first propose an improved PDAM module in the backbone network to form a new feature extraction network to make the learning of the network more consistent with the human cognitive process, so as to improve the model's ability to extract and distinguish subtle features. In the neck network, we apply an improved BF-FPN module to the segmentation task, followed by designing a BHDF-FPN module in the detection task. The BF-FPN and BHDF-FPN modules are tailored to meet the distinct requirements of each task for features after decoupling, thus enhancing the model's versatility in complex scenarios.

The BHDF-FPN model is an extension of the BF-FPN, designed to further enhance feature fusion. Through multilayer dilated convolution superposition and bidirectional fusion, the BHDF-FPN generates an output feature map that effectively balances both semantic information and fine-grained details, thereby reducing the likelihood of missed detections, which can greatly benefit detection tasks. In contrast, for segmentation tasks, it is necessary to perform pixel-level segmentation based on the detection results. Although semantic information needs to be strengthened based on the feature pyramid network (FPN [51]), the essence of the task is more focused on the demand for pixel details. However, using BF-FPN in the segmentation task can preserve the details of the feature map and obtain a more accurate segmentation edge contour and a smoother segmentation curve closer to the target edge, which is more suitable for meeting this requirement (Note: This will be verified in Sections 4.4 and 4.3.2 in terms of predicted results and features, respectively), and thus, the decoupling architecture design employs both modules. Fig. 2 shows the network architecture.

The decoupling information flow is shown in algorithm 1.

During the training stage, the MTFDN architecture first uses PDAM to give channel attention and spatial attention different importance levels at different stages, and then extracts features from input images through residual networks to generate four different feature layers, denoted as $C1 \sim C4$. The resulting feature maps are then processed using the BHDF-FPN and BF-FPN to perform different tasks.

The BHDF-FPN enables feature map enhanced extraction under the “humanized pupillary field of vision” and bidirectional fusion to form a new feature combination as $P1 \sim P4$. Following upsampling, the resulting feature layer $P4$ is directly transformed into layer $P5$. The RPN utilizes the feature pyramid [$P1, P2, P3, P4, P5$] to generate proposal boxes. RoI Align is used to map the proposal boxes to the corresponding feature maps of the input image. The resulting features are then processed through convolutional operations with 1024 channels using 7×7 and 1×1 convolutional kernels. These features are further processed using a fully connected layer and a softmax operation to obtain the classification labels and calculate the classification loss. Additionally, the fully connected layer and reshape operation are used to obtain the positional offsets for each proposal box, which are used to calculate regression losses.

The BF-FPN generates the feature pyramid [$Q1, Q2, Q3, Q4$] following the bidirectional fusion of the extracted stage features. The output feature pyramid is put not into the RPN but into the segmentation branch RoI Align together with the candidate box generated by the BHDF-FPN through the RPN. Continuous convolution is performed, starting with a 3×3 convolution operation with 256 channels, followed by a 2×2 transposed convolution to double the size of the feature map. Finally, the predicted segmentation result of the corresponding category in the proposal box is compared with the true value to calculate the segmentation loss.

During the prediction stage, the source of the proposal boxes is different from the training phase. The prediction boxes and category information of the final output of the detection branch detector are input

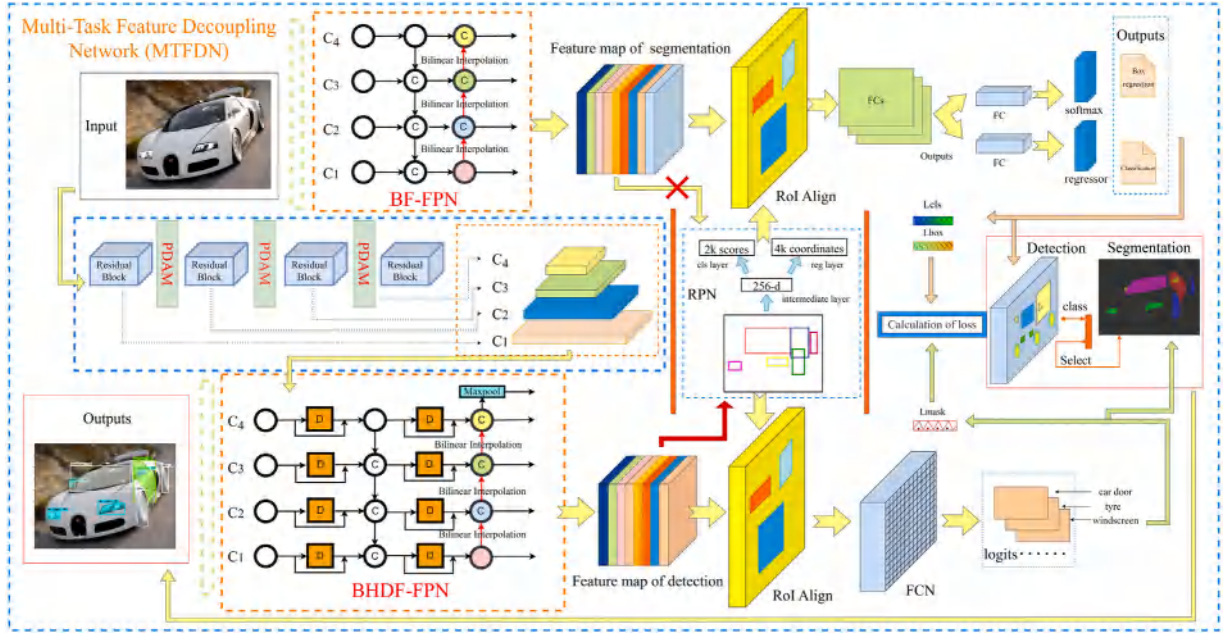


Fig. 2. Architecture of the MTFDN.

Algorithm 1: Information decoupling flows

Input: $Label_r$: The labels for real categories ;
Input: $BBox_{gd}$: The real target boxes ;
Input: $Backbone$: Feature extraction network (ResNet+PDAM) ;
Input: BHDF – FPN: The feature pyramid network for detection task ;
Input: BF – FPN: The feature pyramid network for segmentation task ;

Output: L_{MTFDN} : The training loss of MTFDN framework ;
Output: $predict_{bb}$: The predicted target box;
Output: $predict_{mask}$: The predicted segmentation mask;

- 1 The picture goes through *Backbone* to extract the feature f ;
- 2 The f is fed into BHDF – FPN and BF – FPN respectively to generate detection task feature $f_{det}^{(d)}$ and segmentation task feature $f_{seg}^{(d)}$;
- 3 **if train then**
- 4 **if detectiontask then**
- 5 $(proposal_{det}, loss_{rpn}) \leftarrow RPN(f_{det}^{(d)});$
- 6 $f_{det,r}^{(d)} \leftarrow RoI_{det}(proposal_{det}(f_{det}^{(d)}));$
- 7 $(Label_p, BBox_{reg}) \leftarrow Soft(MLP(f_{det,r}^{(d)}));$
- 8 $loss_{bb} \leftarrow CE(BBox_{reg}, BBox_{gd});$ Calculate $loss_{bb}$;
- 9 $loss_{cls} \leftarrow CE(Label_p, Label_r);$ Calculate $loss_{cls}$;
- 10 **if segmentationtask then**
- 11 $f_{seg,r}^{(d)} \leftarrow CNNs(RoI_{seg}(proposal_{det}(f_{seg}^{(d)})));$
- 12 $loss_{seg} \leftarrow CE(mask_{gd}, Select_{Label_p}(FCN(f_{seg,r}^{(d)})));$
- 13 **end**
- 14 **end**
- 15 **else**
- 16 **if validation then**
- 17 $proposal_{det} \leftarrow RPN(f_{det}^{(d)});$
- 18 $(predict_{bb}, Label_p) \leftarrow Pre(RoI_{det}(proposal_{det}(f_{det}^{(d)})));$
- 19 $f_{seg,p}^{(d)} \leftarrow CNNs(RoI_{seg}(predict_{bb}(f_{seg}^{(d)})));$
- 20 $predict_{mask} \leftarrow Select_{Label_p}(FCN(f_{seg,p}^{(d)}));$
- 21 **end**
- 22 **end**
- 23 **if train then**
- 24 **return** $L_{MTFDN} = loss_{rpn} + loss_{bb} + loss_{cls} + loss_{seg}$
- 25 **else**
- 26 **if validation then**
- 27 **return** $predict_{bb}, predict_{mask}$
- 28 **end**
- 29 **end**

into the split branch RoI Align together with the feature diagram of the BF-FPN output. For each category, a corresponding mask is generated. The mask that corresponds to the category information predicted by the detection branch is then selected, and bilinear interpolation is employed to scale the mask to the same size as the prediction boxes. Finally, the mask is placed in the corresponding region of the original image to yield the segmentation image output.

Although the series of segmentation operations is semantic segmentation, because there is only one category in each candidate box, it is equivalent to instance segmentation, similar to the original Mask RCNN. In addition, MTFDN can be integrated into other top-down two-stage multi-task instance segmentation algorithms, which will be proved in Section 4.6.

3.2. Module design: Parallel dynamic attention module (PDAM)

A convolutional block attention module (CBAM [52]) is a lightweight universal module that can be integrated into various convolutional neural networks for end-to-end training. For a feature map generated by a CNN, a CBAM calculates the attention map from the channel and space dimensions, and then multiplies the attention map with the input feature map to carry out adaptive feature learning.

Based on our test results, the CBAM's processing of feature weights through dual multiplication exerts minimal impact on the resultant feature image and confers marginal benefits to the overall feature extraction process for the vehicle components.

The function of the channel attention module is to increase the effective channel weights and reduce the invalid channel weights. The spatial attention module assigns different importance degrees to different regions of the feature image in space to increase the area of interest and suppress useless areas.

The channel attention module calculation function is defined, and its characteristic weight function is:

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (1)$$

where AvgPool() is the average pooling function; MaxPool() is the maximum pooling function; MLP() is the multilayer perceptron function; and $\sigma()$ is the sigmoid activation function.

Available:

$$F' = M_c(F) \otimes F \quad (2)$$

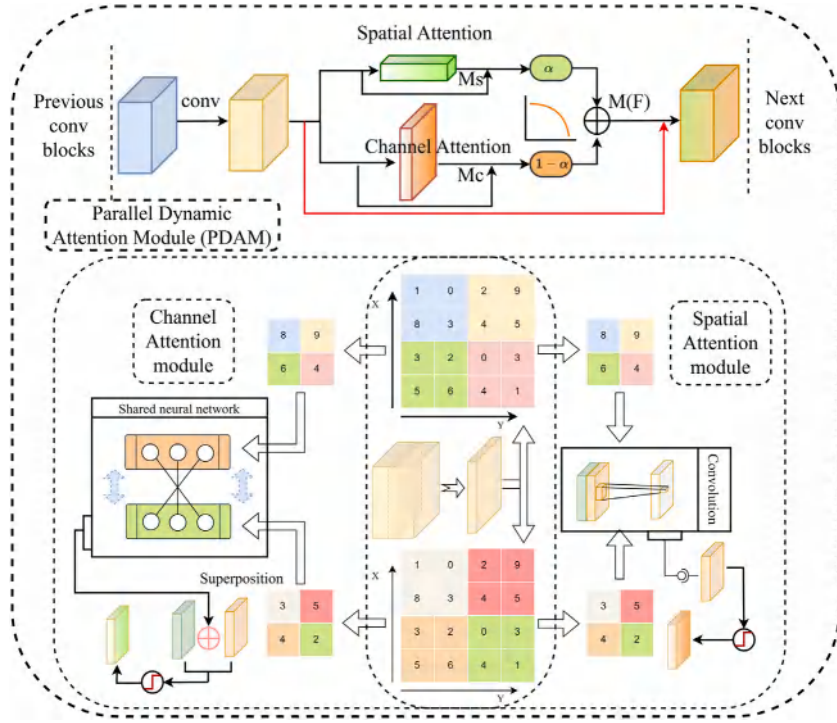


Fig. 3. PDAM structure diagram.

where $\otimes$ represents the multiplication of corresponding elements, and F' represents the feature image after passing through the attention module.

Let $g^{7 \times 7}[]$ be a 7×7 convolution kernel function, define the spatial attention module calculation function, and its characteristic weight function $M_s(F')$ is:

$$M_s(F') = \sigma(g^{7 \times 7}[\text{AvgPool}(F'); \text{MaxPool}(F')]) \quad (3)$$

In addition:

$$F'' = M_s(F') \otimes F' \quad (4)$$

where F'' is the feature image after passing through the channel attention.

Furthermore, it is well known that when the initial training performance is poor, if we pass information through a channel attention mechanism and then a space attention mechanism in the same training stage, multiplying two modules in series will cause the model to tend to focus on unrelated features and regions. But in the context of vehicle components detection where the feature differences between parts are relatively small, human-like sequential learning of features is crucial for distinguishing the subtle feature variances between vehicle components. This requires dynamic adjustments during the training stage. To address this, we connect the channel and spatial attention modules in parallel and then multiply them by coefficients with different influence weights during different training stages while ensuring that their sum is always 1. This helps adjust the influence weights between the modules in different training stages. The output of the two modules is then mapped to a unified size and summed, and the sigmoid function is used to obtain the final attention map $M(F)$ within the range of 0 to 1. Element-by-element summing is an effective way to integrate and protect information from the previous layer in terms of information flow. This enables the network to utilize information from the complementary channel and space branches without losing any information during the forwards phase. In the back phase, the gradients are evenly distributed across all inputs, leading to effective training. In contrast, adopting an element-by-element product may lead to a large gradient being assigned to a small input during the dynamic

adjustment process, making it challenging for the network to converge and resulting in poor performance. Therefore, we use the element-by-element sum in our proposed method, which can be expressed as follows:

$$M(F) = \sigma(M_c(F) \otimes (1 - \alpha) + M_s(F) \otimes \alpha) \quad (5)$$

where α is the channel influence factor and $\sigma()$ is the sigmoid activation function. Both branch outputs are adjusted to the size of $R^{C \times H \times W}$ before being added.

To achieve the goal of assigning varying degrees of importance to channel and spatial features at different stages of training, we propose an attention mechanism that adapts its attention weight dynamically. Specifically, we increase the attention weight of channels during the initial phase of training to emphasize their relevance (broad learning of general knowledge). In later stages, we shift towards spatial attention to focus on identifying areas of interest (domain related knowledge). This simulates the model training process as the construction process of human learning and cognition: the establishment of human cognition about the world begins with learning a broad range of scattered knowledge to construct a macroscopic cognitive world (corresponding to the model's learning of general knowledge), and then, with the accumulation of learning and experience (corresponding to the model's training process), cognitive abilities gradually deepen (corresponding to the model's performance improvement), becoming more specialized and profound, eventually having their own professional domains (model focusing on predefined category information, discarding irrelevant information). To accelerate the transition between these two attention influences in the mid-term of training, we implement a weight transition strategy. In summary, the mechanism we propose endows the model's learning process with human-like sequentiality, effectively prioritizing channel and spatial features according to cognitive logic during different training stages.

The following formula satisfies the above description:

$$\alpha = \frac{1}{2} \sin\left(\frac{\text{epoch}}{\text{epochs}} \pi - \frac{\pi}{2}\right) + \frac{1}{2} \quad (6)$$

where epoch indicates the current training cycle and epochs indicates the total training cycle.

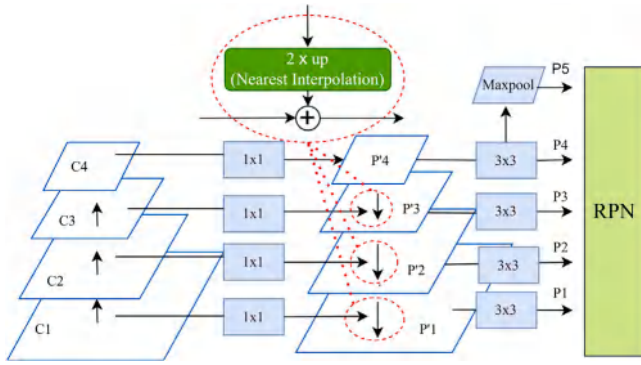


Fig. 4. FPN structure diagram.

The parallel dynamic attention module (PDAM) structure is shown in Fig. 3.

Incorporating the PDAM into the feature extraction network has the potential to enhance the effectiveness of features in both spatial and channel dimensions, improved channel and spatial connection of features, suppress the flow of irrelevant information, and facilitate the extraction of target-specific features. In accordance with the methodology described in Woo et al. [52], this investigation applies the PDAM to all ResNet50 feature extraction convolutional residual blocks. The previous residual block output undergoes convolution to generate feature image F , which serves as input to the attention module. Following spatial and channel attention module processing, the outputs are multiplied respectively by the corresponding influence coefficients, resized to a uniform dimension, and added to produce the attention map. The attention map is subsequently multiplied by the input feature map to enable adaptive feature learning.

3.3. Module improvement: Bidirectional fusion feature pyramid network (BF-FPN)

Considering the distinct segmentation task features described in Section 3.1, we propose an improved feature fusion network, named the bidirectional fusion feature pyramid network (BF-FPN), to boost the accuracy of vehicle component segmentation. The BF-FPN incorporates the fusion concept from the literature [53–55] into a FPN [51] framework. By leveraging this novel approach, we aim to achieve superior segmentation performance in the vehicle domain.

The FPN structure is shown in Fig. 4. The input image is first passed through ResNet to obtain the feature map of different scales from C1 to C4. Then, taking P3 as an example, the output obtained by convolving C3 using a 1×1 kernel is superimposed with the feature map obtained by upsampling, ultimately resulting in the generation of P3.

The BF-FPN model enhances the downwards fusion process of the FPN and introduces an upwards fusion process to enhance the ability to express the boundary of the target. As illustrated in Fig. 5, the BF-FPN structure comprises three key processes: an overall feature map forwards propagation process, a top-down fusion process, and a bottom-up fusion process.

In this study, we propose a modification to the bidirectional fusion process in the FPN. In comparison to PANet [53], although our BF-FPN has a similar bidirectional fusion process, we have made greater efforts to retain details due to the need for details at the mask boundary (the high frequency part of the image) of the segmentation task. Specifically, we no longer employ the nearest interpolation method for upsampling, but instead opt for bilinear interpolation. This is because the nearest interpolation destroys the gradual relationship between pixel values in the original image when generating new images. Therefore, we opt for bilinear interpolation. Meanwhile, we replace the direct addition of multidimensional features with a concatenate operation, as it is

not conducive to the subsequent segmentation task of full convolution. Furthermore, we have applied appropriate 3×3 convolution operations after feature fusion in adjacent layers to prevent feature cluttering, minimize over-smoothing and coarseness of segmentation target masks in high-frequency areas, and reduce misclassification of a large number of edge pixels. The primary intention behind these design choices is to preserve pixel-level detail information.

While preserving detail information as much as possible, the bottom-up fusion process involves integrating low-level feature maps containing detailed information with high-level feature maps rich in semantic information, replacing the original high-level feature maps, thereby ensuring the integrity of segmentation for large objects. This process involves downsampling the lower-level feature map to match the size of the upper layer, followed by concatenate superposition with the forwards-generated feature map using a transverse connection. This process is a bottom-up superposition process, which also requires channel matching and a 3×3 convolution to remove aliasing.

In the original RPN, nine anchors of different sizes are selected on the output feature map. In our study, we propose a multiscale feature map and establish a dimension correspondence between the feature map of each layer. Specifically, the sizes of the anchors corresponding to [Q5, Q4, Q3, Q2, Q1] are $[512 \times 512, 256 \times 256, 128 \times 128, 64 \times 64, 32 \times 32]$, and we select three ratios of [2:1, 1:2, 1:1] for each size. As a result, a total of 15 anchor types are employed in our proposed modified BF-FPN. To determine the RoI pooling layer input strategy, we refer to the method used in FPN.

3.4. Module design: Bidirectional hybrid dilated fusion feature pyramid network (BHDF-FPN)

Fig. 4 illustrates that the FPN upsampling from P4 to P3 and P3 to P2 can cause irreversible image information loss. Although simple feature fusion techniques are employed on each FPN layer feature map to reduce the information loss, small features can still be easily ignored. This issue persists in the BF-FPN.

To achieve robust vehicle component detection in complex scenes with poor visibility during night-time, visual deformation of vehicle components due to changes in scene location, image blur caused by equipment during shooting, and other challenging identification conditions, it is imperative to enhance feature fusion further in detection tasks. To this end, we propose a novel module by leveraging the advantages of dilated convolution and following the hybrid dilated convolution (HDC [56]) construction principle. Specifically, we apply multilayer dilated convolution to the BF-FPN, resulting in the proposed BHDF-FPN. The BHDF-FPN is designed to capture and fuse more image information within the “humanoid pupillary field of vision”, which leads to a significant improvement in the detection accuracy.

In this study, we explore the benefits of dilated convolution for increasing the receptive field without losing information due to pooling operations. By controlling the convolution kernel rate, we can obtain receptive fields of varying sizes. The dilated convolution modules in Fig. 6 consist of 3×3 convolution kernels, and when the rate is set to 1, the receptive field is 3×3 , while a rate of 2 corresponds to a receptive field of 7×7 . Accordingly, we construct the BHDF-FPN module by incorporating dilated convolution into the BF-FPN module in the upper section. To avoid the grid effect and capture more features under the receptive field, we follow the HDC construction principle and use dilated convolution, whose multilayer expansion rate common divisor is not greater than 1, to perform superposition. This helps to obtain denser data under the multilayer receptive field while retaining the spatial features of the image with limited computational resources. The feature graphs fused with dilated convolution superposition in advance are used for upper and lower sampling in the top-down and bottom-up fusion sampling process, which effectively reduces the loss of image information caused by sampling operations. In addition, inspired by the Receptive Fields (RFs [57,58]) of the human visual system, it is

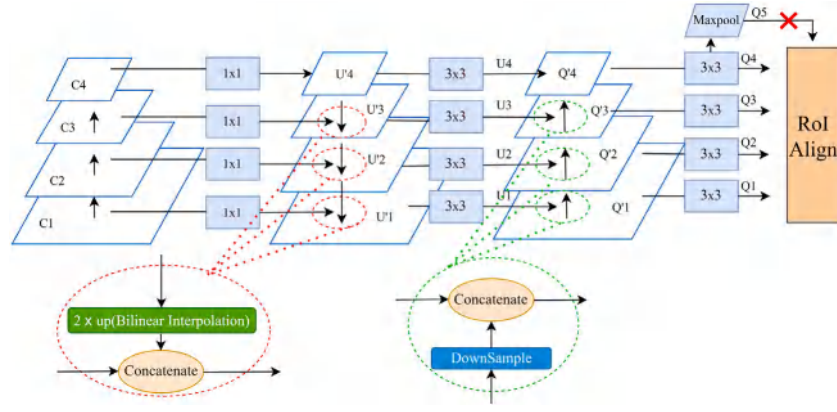


Fig. 5. BF-FPN structure diagram.

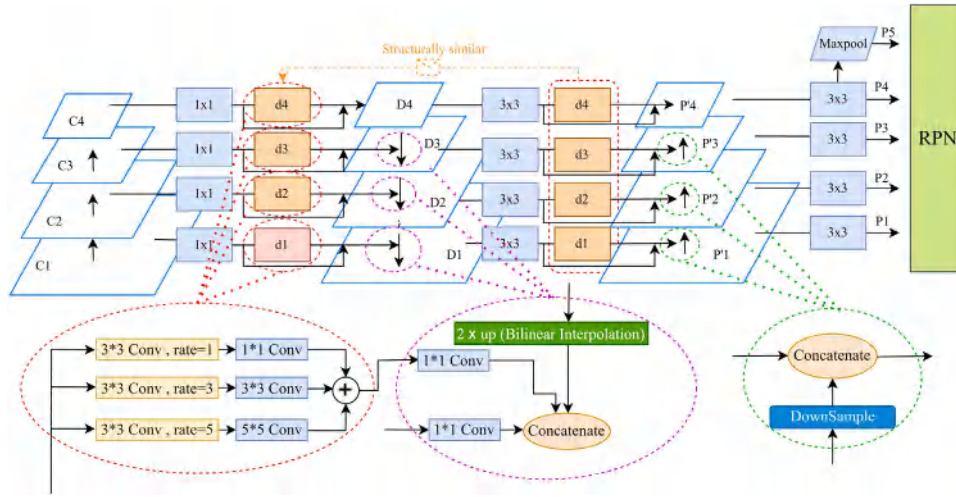


Fig. 6. BHDF-FPN structure diagram.

Table 1

The experimental equipment and training parameters.

Equipment/Parameter	Version number/Specific value
Display card	NVIDIA RTX 3090
Operating system	Ubuntu18.04 LTS
Programming environment	Python 3.7, Pytorch 1.8
Parameter optimization method	SGD
Learning rate	0.002
Learning rate decay ratio	10%
Maximum training rounds	20
Channel influence factor α	$\alpha = \frac{1}{2} \sin\left(\frac{epoch}{epochs} \pi - \frac{\pi}{2}\right) + \frac{1}{2}$

desirable to position the target area as close as possible to the centre of the receptive field. In this regard, this study refers to the construction of the RFB module [59], providing the computer with a field of view that, similar to the human eye pupil, enhances the receptive field. Specifically, the scheme involves the parallel connection of three output feature maps of varying sizes and receptive fields, thereby enhancing the robustness of the model to small-scale spatial displacement and improving the accuracy of the detection network results.

The BHDF-FPN structure diagram is shown in Fig. 6. If only a large rate is adopted, only the detection of large objects can be improved. We propose a dilated convolution superposition structure of three layers with a common divisor expansion rate not greater than 1 to address this limitation. To illustrate, D1 is taken as an example, where C1 is convolved using three kinds of 3×3 dilated convolutions (rate=1, rate=3, rate=5) to obtain three kinds of feature graphs and then using a

convolution kernel (1×1 , 3×3 , 5×5) to provide different weights to each feature graph layer to form a receptive field similar to the human eye pupil, as shown in Fig. 7. The resultant feature map is then obtained via a parallel connection operation. The output of d1 and C1 both undergo 1×1 convolution, and their respective outputs are superimposed with the upscaled outputs of D2 to obtain D1. D1 fuses feature graphs of different scales while reducing information loss due to up- and downsampling. The subsequent steps follow this pattern.

4. Experiment and analysis

4.1. Experimental platform and parameter settings

To ensure the reliability of the comparative ablation experiment and ensure that the parameters in each model are close to the optimal value, several training tests were conducted for each model before the comparative experiment, and a set of super parameters were selected as a compromise for the comparison ablation experiment. The devices and settings shown in Table 1 are used in Section 4.3.2.

4.2. Experimental dataset

In this study, we establish two kinds of datasets for internal and external factors, which also have different vehicle component distributions. Specifically, one dataset comprises long-range pictures of vehicles in various scenarios collected from the internet, which is referred to as VSD in this paper. The other dataset comprises close-up vehicle

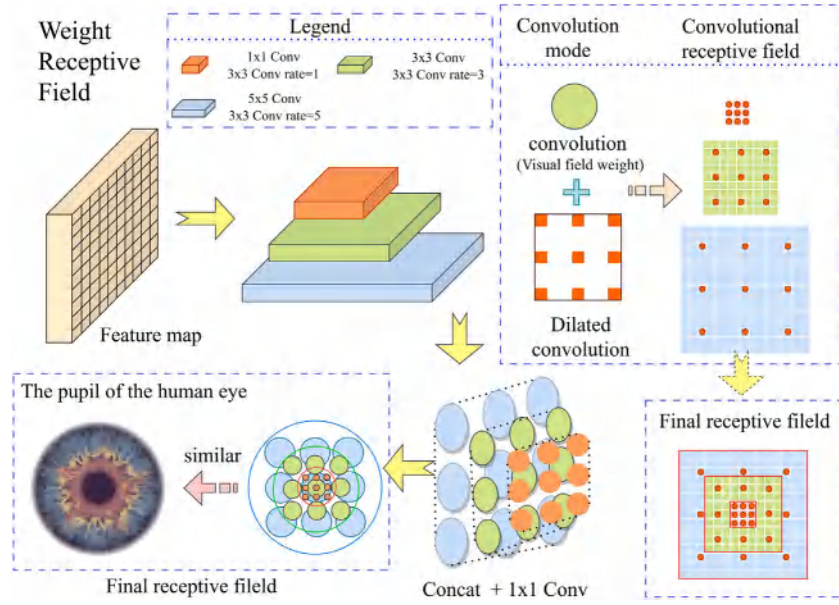


Fig. 7. Weight receptive field.

component images taken by an insurance company at the scene of vehicle insurance loss determination represented as a local vehicle form, which is denoted as VCD throughout this paper.

We annotated the VSD and VCD datasets using LabelMe and divided them into training sets and test sets. VSD contains 1283 training and 361 test images, while VCD contains 6694 training and 1674 test images. The VSD images originated online and are captured from streets, shopping malls, beaches, parks, highways, stations, gas stations, and other complex daily life scenes. In addition, it includes weak light environments such as night and morning, and high exposure environments such as hot sun and mirror reflection as far as possible. The dataset covers five main categories of vehicle components, namely, car doors, car lights, rearview mirrors, tires, and windcreens. In contrast, the VCD images are primarily partial vehicle components taken by insurance companies to determine the extent of vehicle loss. The dataset includes 59 vehicle component categories, such as the exterior roof plate, reverse mirrors, underwire, trunk cover, and rear bumper skin. The VSD images conform to a balanced distribution of categories and depict panoramic views of the vehicles, thereby making them more suitable for road vehicle detection applications. Conversely, the VCD images are imbalanced in terms of category distribution and mainly depict partial views of the vehicles, making them more relevant for vehicle loss determination.

To sum up, VSD is a vehicle perspective dataset with a small number of marked vehicle component categories and large differences among categories, so the main factors affecting detection and segmentation are the external factors summarized in Section 1. VCD is a close-range vehicle dataset with many marked vehicle component categories and small differences among categories, so the main factors affecting detection and segmentation are the internal factors summarized in Section 1. Therefore, in this study, VSD is used to reflect the improvement of external factors in the comparison experiment, and VCD is used to reflect the improvement of internal factors in the comparison experiment.

4.3. Experiment

4.3.1. Performance indicators

In this study, the mean average precision (mAP) and mean average recall (mAR), which are commonly used in object detection segmentation models, are used to evaluate the model as a whole.

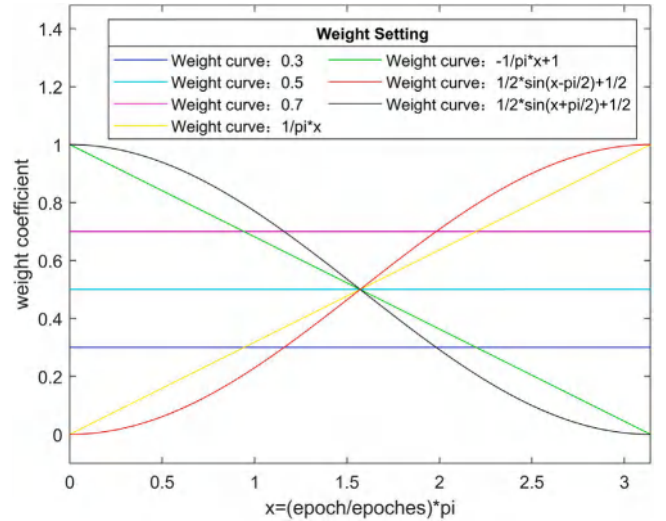


Fig. 8. Weight curves used to compare experimental settings.

The mAP is defined as:

$$mAP = \frac{1}{C} \sum_{n=1}^C \int_0^1 P_n(R_n) dR_n \quad (7)$$

where C is the number of categories of components, R_n is the recall rate of class n , and $P_n(R_n)$ is the precision of class n when the recall rate of class n is R_n . mAP is used to quantitatively evaluate the model training effect, which can reflect the training model performance. In detection, the parameters correspond to the IoU of Bbox, and in segmentation, the parameters correspond to the IoU of pixel-level Segm.

The mAR is defined as:

$$mAR = \frac{2}{C} \sum_{n=1}^C \int_{0.5}^1 recall(o) do = \frac{2}{jC} \sum_{n=1}^C \sum_{i=1}^j \max(IoU(gt_i) - 0.5, 0) \quad (8)$$

where C is the number of categories of components, j is the number of overlaps between all ground truth bounding boxes in each picture and the detection result bounding boxes that are closest to ground truth bounding boxes. The IoU is the result of dividing the intersection

Table 2
PDAM comparison and ablation experiments.

		Method				$AP_{50}(\%)$	
	Mask RCNN	FPN	CBAM	PDAM	α	Bbox	Segm
Single channel	✓	✓	–	–	–	87.4	86.4
			✓	–	–	88.2	87.7
			–	✓	0.3	87.9	86.7
			–	✓	0.5	88.5	87.6
			–	✓	0.7	87.5	86.4
			–	✓	$1/\pi * x$	86.1	85.3
			–	✓	$-1/\pi * x$	85.9	84.7
			–	✓	$1/2 * \sin(x + \pi/2) + 1/2$	88.1	87.5
			–	✓	$1/2 * \sin(x - \pi/2) + 1/2$	89.6	88.6

* Note: In this study, we compared three static values of α . Our results indicate that when the spatial attention and channel attention weights are set equally, the results are better, which is equivalent to the BAM method. However, when the weights of $\alpha = 1/\pi * x$ and $\alpha = -1/\pi * x$ are set, the training results may occasionally converge poorly, and the mAP results may fluctuate. Moreover, we compared the weight settings of the two trigonometric function forms. Our findings suggest that gradually reducing the channel attention weight and increasing the spatial attention weight during the training process is more effective. This can be interpreted as focusing on optimizing the effective features in the early stage and increasing their weight and subsequently identifying areas of interest based on this foundation in the later stage. This is also the best way to compare the previous settings. Therefore, $\alpha = 1/2 * \sin(x - \pi/2) + 1/2$ is adopted in this study.

** Note: where, $x = \frac{epoch}{epochs} \pi$.

Table 3
Comparisons with different variations of FPN.

Algorithm framework	Module						$AP_{50}(\%)$			
	FPN	PANet	Bi FPN	NAS-FPN	BF-FPN	BHDF-FPN	VSD		VCD	
							Bbox	Segm	Bbox	Segm
Mask RCNN	✓						87.4	86.4	39.3	37.9
		✓					89.0	88.3	41.7	40.9
			✓				89.6	88.7	42.6	41.0
				✓			88.2	87.5	41.2	40.4
					✓		88.7	89.8	41.5	42.7
						✓	90.7	89.1	43.3	41.4

Table 4
Based on VSD, compared with the single (double) channel network replacing the FPN with the BHDF-FPN.

Remarks		Method				$AP_{50}(\%)$		$AP_{50:95}(\%)$		$AR_{50:95}^{max Dets=100}(\%)$	
		FPN	BF-FPN	BHDF-FPN	PDAM	Bbox	Segm	Bbox	Segm	Bbox	Segm
Single channel	*	✓	–	–	–	87.4	86.4	56.0	59.5	64.6	67.0
	o	–	–	✓	✓	91.5	89.8	61.0	61.7	67.8	67.9
	•	–	–	✓	✓	91.6	89.7	61.1	61.6	67.5	68.0
Dual channel	★	–	✓	✓	✓	91.9	90.5	61.1	62.9	67.6	68.4

* NOTE: Both “o” and “•” exhibit identical feature flow directions. The purpose of comparing these two is to eliminate the impact of simple channel factors, which refers to potential random errors between the feature outputs generated by the same BHDF-FPN (“o”) and the two feature graphs generated by two identical BHDF-FPNs (“•”) on experimental outcomes. Fig. 9 depicts the network architecture of both “o” and “•”. The comparative test results are presented in the table above. When influencing factors other than the number of channels remain the same, the performance of “o” is observed to be comparable to that of “•”. Therefore, the influence of the simple channel factor on the experimental results can be excluded.

of the two regions (ground truth bounding boxes and detection result bounding boxes) by the union of the two regions. For segmentation, the parameters used for calculation correspond to the relevant parameters of the mask. AR is the evaluation standard of the missed detection of the detector. mAR represents the average of AR across all categories.

4.3.2. Ablation experiment comparisons

To validate the effectiveness of the proposed PDAM and the rationality of its parameter settings, we conducted a comparative experiment by inserting a CBAM and PDAM into the Mask RCNN. Additionally, we set the weight curve, as presented in Fig. 8, when applying the PDAM for comparison purposes. This analysis enables us to evaluate the impact of our proposed mechanism on the performance of the network in terms of detection and segmentation performance. The ablation experiment results are shown in Table 2.

Subsequently, we aim to validate the efficacy of the proposed BHDF-FPN and BF-FPN architectures in enhancing object detection segmentation model performance. To this end, we conduct a comparative evaluation by substituting the FPN module in the Mask RCNN with our proposed BHDF-FPN and BF-FPN architectures. Meanwhile, other FPNs are compared. The ablation experiment results are shown in Table 3.

Then, we aim to evaluate the impact of the proposed modifications to the overall network architecture on the object detection and segmentation results. Specifically, we investigate the effect of adopting the dual-channel BHDF-FPN and BF-FPN architecture on the network performance by replacing the FPN module with the BHDF-FPN in the detection task and the BF-FPN in the segmentation task to meet the distinct requirements of each task for features after decoupling, resulting in superior network boundary segmentation performance and feature extraction ability. The comparative effects of the BHDF-FPN and BF-FPN dual-channel network, the BHDF-FPN replacing the FPN single-channel network and the baseline Mask RCNN are shown in Tables 4 and 5.

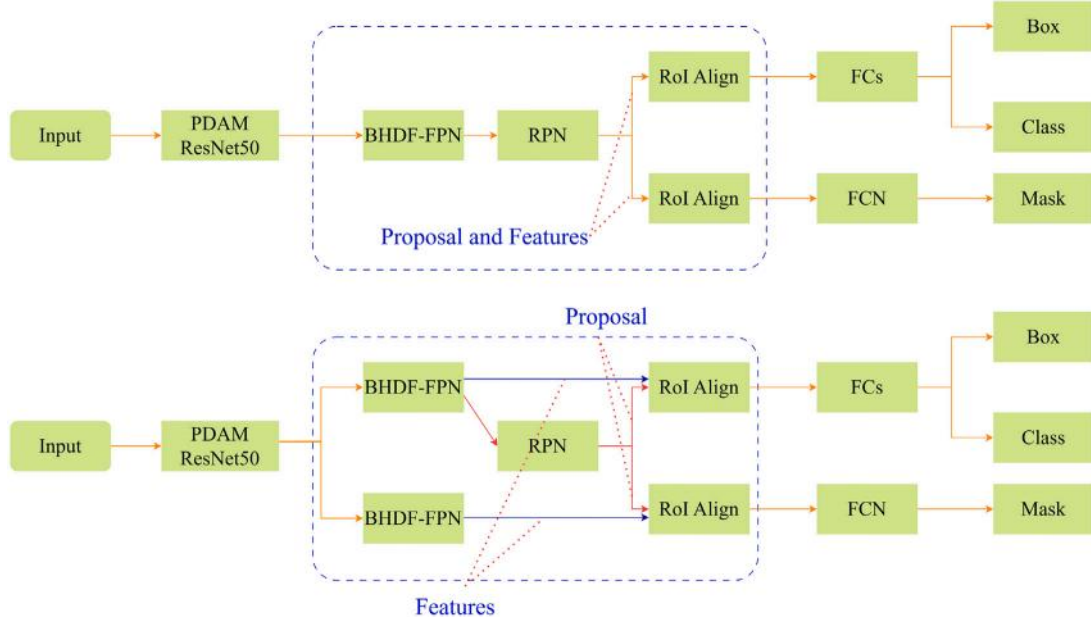
To ensure detection performance comparable to the BHDF-FPN while mitigating the feature aliasing phenomenon and detail information loss caused by high feature fusion, the BF-FPN is employed to provide segmentation feature maps, with the BHDF-FPN serving as the detection task. This approach enhances segmentation performance, as depicted in Fig. 10. Notably, the dual channel decoupling method outperforms alternative approaches in accurately delineating the boundaries of detected targets.

Table 5

Based on VCD, compared with the single (double) channel network replacing the FPN with the BHDF-FPN.

	Remarks	Method				$AP_{50}(\%)$		$AP_{50:95}(\%)$		$AR_{50:95}^{max Dets=100}(\%)$	
		FPN	BF-FPN	BHDF-FPN	PDAM	Bbox	Segm	Bbox	Segm	Bbox	Segm
Single channel	*	✓	–	–	–	39.3	37.9	28.3	27.9	60.7	58.9
	◦	–	–	✓	✓	46.7	44.3	33.2	33.1	68.9	67.1
	•	–	–	✓	✓	46.6	44.2	33.0	33.1	69.0	67.3
Dual channel	★	–	✓	✓	✓	46.8	45.7	33.0	34.2	69.1	68.4

* NOTE: Same as Table 4.

**Fig. 9.** BHDF-FPN single/dual channel structure diagram and its information flow direction.**Fig. 10.** The BF-FPN and BHDF-FPN (right) prediction effect compared with that of the BHDF-FPN (left).**Table 6** $AP_{50}(\%)$ table with baseline Mask RCNN+FPN for each component on VSD.

Network		Car door		Car light		Rearview		Tyre		Windscreen	
		Bbox	Segm	Bbox	Segm	Bbox	Segm	Bbox	Segm	Bbox	Segm
Single channel	Mask RCNN+FPN	90.4	87.7	74.1	73.4	87.7	88.1	95.9	96.0	88.7	86.5
Dual channel	(Textual Method) MTFDN	96.6	92.2	76.8	76.7	93.3	93.5	99.0	99.0	94.0	92.1

To date, MTFDN has achieved significant advancements in both detection and segmentation accuracy owing to its distinctive dual-channel decoupling architecture comprising a BHDF-FPN and BF-FPN. When evaluating this approach on VSD, focusing on five specific vehicle component categories, the results shown in Table 6 are obtained, demonstrating notable improvements compared to the baseline Mask RCNN approach. When evaluating the MTFDN model on VCD, visual analysis of Figs. 11 and 12 reveals a significant improvement in

the detection and segmentation performance of most components in comparison to the baseline model. In addition, it can be seen from the figures that the variation trend of segmentation performance and detection performance is highly coincident, which is the embodiment of segmentation before detection. Therefore, our method carries out targeted decoupling design on the feature extraction methods of detection and segmentation, which not only maximizes detection performance but also uses fine-grained features to meet the needs of segmentation.

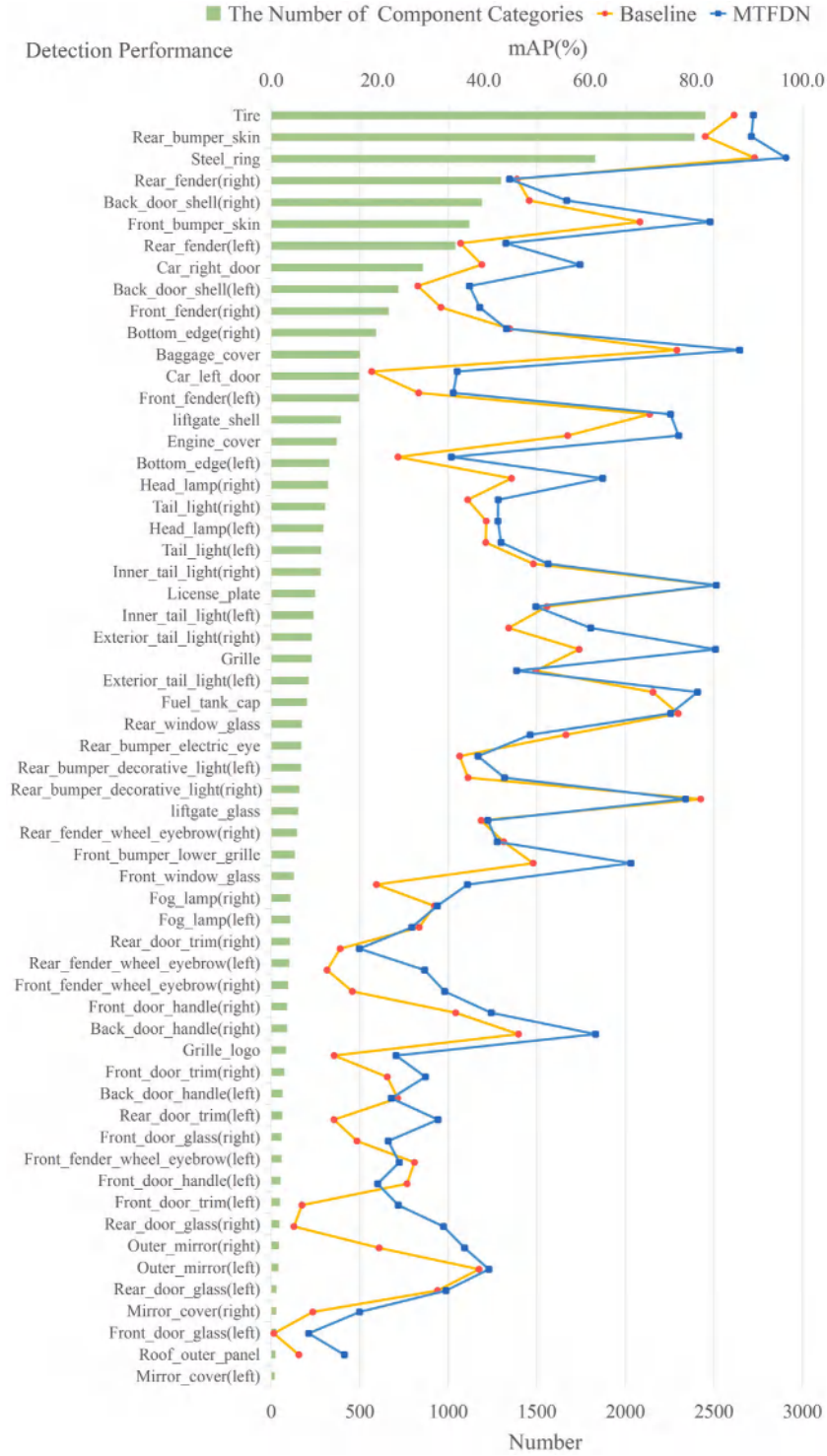


Fig. 11. Comparison with the baseline Mask RCNN+FPN for each component of the detection performance on VCD.

Then we use the primary and secondary relationship between tasks in the network prediction post-processing part, which realizes the overall performance improvement and conforms to the objective law.

In addition, compared with a single-channel FPN, the MTFDN framework incorporates two variants, namely, the BHDF-FPN and BF-FPN, as substitutes for the conventional FPN model and updates the segmentation task feature route, which does not extend the overall network path length. Meanwhile, MTFDN utilizes PDAM updates the real-time weights concerned of the training focus channels and regions, which does not increase significantly the number of parameters. So the

detection time does not increase significantly compared to the original network, ensuring the detection rate. The ablation experiment results are shown in Table 7.

Table 8 presents the results of the module ablation experiments. It is evident from the individual ablation of each module that the separate contributions of each module are as expected. After the reasonable embedding of each module into the decoupled framework, there is a significant improvement in the detection and segmentation performance. The similar results of ablation experiments 5 and 6 indicate that the transformation of the driving relationship of the decoupled

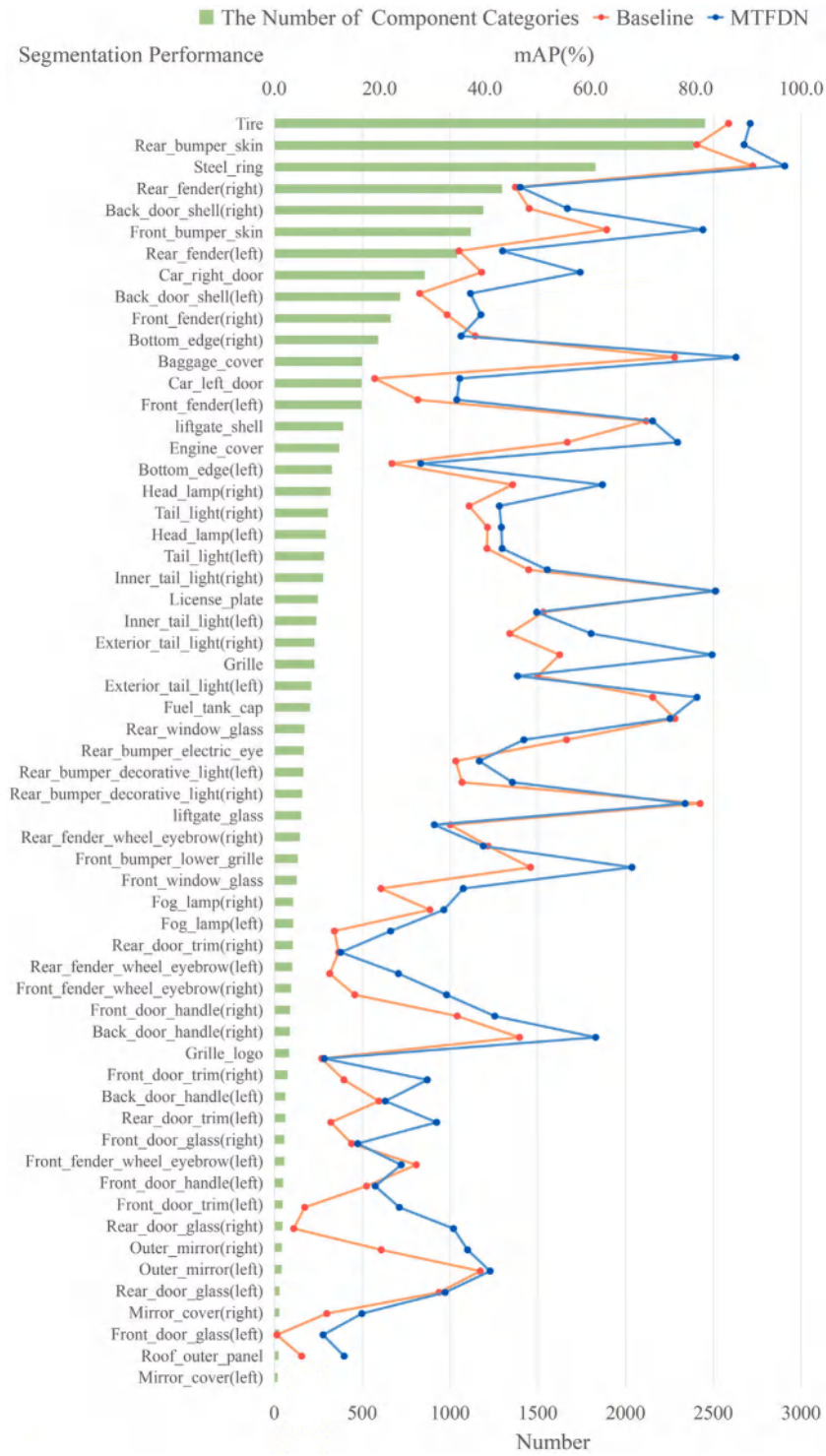


Fig. 12. Comparison with the baseline Mask RCNN+FPN for each component of the segmentation performance on VCD.

Table 7

Detection time comparison.

	Algorithm framework	MTFDN	FPN	BF-FPN	BHDF-FPN	PDAM	Detection rate(s)	
							VCD	VSD
Single channel	Mask RCNN	–	√	–	–	–	0.2959	0.1501
			–	√	–	–	0.3071	0.1556
			–	–	√	–	0.3138	0.1598
			√	–	–	√	0.3056	0.1547
Dual channel	Mask RCNN	√	–	√	√	√	0.3264	0.1631

Table 8

Ablation experiments.

	Method						$AP_{50}(\%)$			
	Mask RCNN	FPN	PDAM	BF-FPN	BHDF-FPN	Channel	VCD		VSD	
							Bbox	Segm	Bbox	Segm
Ablation 1		✓	–	–	–	Single	39.3	37.9	87.4	86.4
Ablation 2		–	✓	–	–	Single	42.1	41.6	89.6	88.6
Ablation 3		–	–	✓	–	Single	41.5	42.7	88.7	89.8
Ablation 4	✓	–	–	–	✓	Single	43.3	41.4	90.7	89.1
Ablation 5		–	✓	–	✓	Single	46.7	44.3	91.5	89.8
Ablation 6		–	✓	–	✓	Dual	46.6	44.2	91.6	89.7
Ablation 7		–	✓	✓	✓	Dual	46.8	45.7	91.9	90.5

Table 9

Comparison with various advanced detection segmentation algorithms on VSD.

Method	Backbone	Neck network	$AP_{50}(\%)$		$AP_{50:95}(\%)$		$AR_{50:95}^{max\ Dets=100}(\%)$	
			Bbox	Segm	Bbox	Segm	Bbox	Segm
(Baseline) Mask RCNN [42]	ResNet-50	FPN	87.4	86.4	56.0	59.5	64.6	67.0
Cascade Mask R-CNN [45]	ResNet-50	FPN	90.2	88.1	60.3	58.4	65.3	62.7
Yolact [29]	ResNet-50	FPN	86.3	87.2	52.8	58.2	–	–
PointRend [60]	ResNet-50	FPN	89.4	89.3	58.2	61.5	64.5	67.1
DCT-Mask [61]	ResNeXt-101	FPN	90.2	88.1	60.1	58.8	65.5	62.3
SOLQ [34]	ResNet-50	–	89.7	87.6	60.0	58.7	65.3	64.8
SparseInst [62]	ResNet-50	–	–	89.8	–	61.0	–	–
Mask Transfuser [46]	ResNet-50	FPN	89.7	88.6	59.7	60.1	64.7	66.3
Yolov7 [63]	⊗	⊗	90.4	88.4	60.0	59.4	–	–
DynaMask [64]	ResNet-50	Dual-level FPN	90.5	89.1	60.1	59.8	63.5	65.8
(Textual Method) MTFDN	ResNet-50+PDAM	BHDF-FPN+BF-FPN	91.9	90.5	61.1	62.9	67.6	68.4

NOTE: Yolov7 [63] does not give backbone a name, this paper replaces it with “⊗”. Similarly, the neck is marked with “⊗”.

Table 10

Comparison with various advanced detection segmentation algorithms on VCD.

Method	Backbone	Neck network	$AP_{50}(\%)$		$AP_{50:95}(\%)$		$AR_{50:95}^{max\ Dets=100}(\%)$	
			Bbox	Segm	Bbox	Segm	Bbox	Segm
(Baseline) Mask RCNN [42]	ResNet-50	FPN	39.3	37.9	28.3	27.9	60.7	58.6
Cascade Mask R-CNN [45]	ResNet-50	FPN	43.1	43.2	31.1	31.6	65.2	64.5
Yolact [29]	ResNet-50	FPN	38.2	36.1	27.4	26.0	–	–
PointRend [60]	ResNet-50	FPN	40.9	40.7	29.8	29.5	63.9	59.7
DCT-Mask [61]	ResNeXt-101	FPN	44.1	43.4	31.9	32.0	65.5	64.9
SOLQ [34]	ResNet-50	–	42.8	42.6	30.8	31.4	65.6	65.1
SparseInst [62]	ResNet-50	–	–	44.3	–	32.8	–	–
Mask Transfuser [46]	ResNet-50	FPN	43.3	43.5	31.3	32.4	64.2	65.8
Yolov7 [63]	⊗	⊗	45.5	44.2	32.2	32.6	–	–
DynaMask [64]	ResNet-50	Dual-level FPN	44.9	44.5	32.1	32.9	64.5	64.2
(Textual Method) MTFDN	ResNet-50+PDAM	BHDF-FPN+BF-FPN	46.8	45.7	33.0	34.2	69.1	68.4

NOTE: Yolov7 [63] does not give backbone a name, this paper replaces it with “⊗”. Similarly, the neck is marked with “⊗”.

framework described in Section 3.1 depends on the decoupled division of labor in the feature extraction part, excluding the influence of purely channel factors. Ablation 7, which adds BF-FPN on top of Ablation 6, leads to an improvement in segmentation performance, further affirming this point.

The comparisons with various advanced detection segmentation algorithms are shown in Tables 9 and 10.

The experimental results demonstrate that compared to the baseline model, our improved algorithm enhances the detection mAP index by 4.5% and the segmentation mAP by 4.1% on VSD. This proves that MTFDN can effectively alleviate the impact of complex scenes and external environments (b in Section 1: External factors) on detection segmentation. We observed an improvement in the detection mAP of 7.5% and in the segmentation mAP of 7.8% on VCD. This proves that MTFDN effectively alleviates the impact of similar component features (a in Section 1: Internal factors) on detection segmentation. Additionally, Tables 9, 10, and 11 demonstrate that the MTFDN framework, while introducing a certain degree of parameter count and computational cost compared to the baseline, still maintains an advantage

in terms of parameter count and computational cost compared to various advanced detection and segmentation algorithms. Furthermore, it exhibits superior performance in terms of detection accuracy and segmentation capability.

4.4. Feature learning class activation diagram display

To intuitively characterize the feature extraction ability of a feature extraction network, the Grad-CAM [65] method was employed for an instance segmentation task focused on detecting and segmenting vehicle components. This method refers to the expression of learning class activation graph features extracted in tasks such as image classification.

However, unlike its application to image classification, object detection, including instance segmentation, requires substantial results processing, such as creating dictionaries containing bounding boxes, labels, and scores and performing the argmax function on scores to identify the highest scoring category, which renders the output non-differentiable. Moreover, the feature maps extracted by backbone are

Table 11

Comparison of real-time performance and model complexity.

Method	Backbone	Neck network	GPU	FLOPs(G)	Params(M)	FPS	
						VCD	VSD
(Baseline) Mask RCNN [42]	ResNet-50	FPN	RTX3090	200.3	43.7	3.379	6.662
Cascade Mask R-CNN [45]	ResNet-50	FPN	RTX3090	276.8	48.5	2.287	4.596
DCT-Mask [61]	ResNeXt-101	FPN	RTX3090	259.7	47.1	2.342	5.289
SparseInst [62]	ResNet-50	–	RTX3090	81.7	31.4	4.018	7.935
Mask Transfuser [46]	ResNet-50	FPN	RTX3090	667.2	54.6	1.351	3.751
Yolov7 [63]	“@”	“@”	RTX3090	178.7	41.2	3.846	7.628
DynaMask [64]	ResNet-50	Dual-level FPN	RTX3090	246.5	46.4	2.523	5.473
(Textual Method) MTFDN	ResNet-50+PDAM	BHDF-FPN+BF-FPN	RTX3090	231.2	45.1	3.064	6.131

NOTE: Yolov7 [63] does not give backbone a name, this paper replaces it with “@”. Similarly, the neck is marked with “@”.

The performance metric “FLOPs” refers to the number of floating point operations required to run a network model once. The performance metric “Params” indicates the number of parameters that need to be learned in the model, which is another important measure of model complexity. The performance metric “FPS” denotes the number of images the network can process per second.

Here, the average FLOPs is calculated using 100 validation images. The FPS is measured on an RTX 3090 using the entire validation set with a batch size of 1.

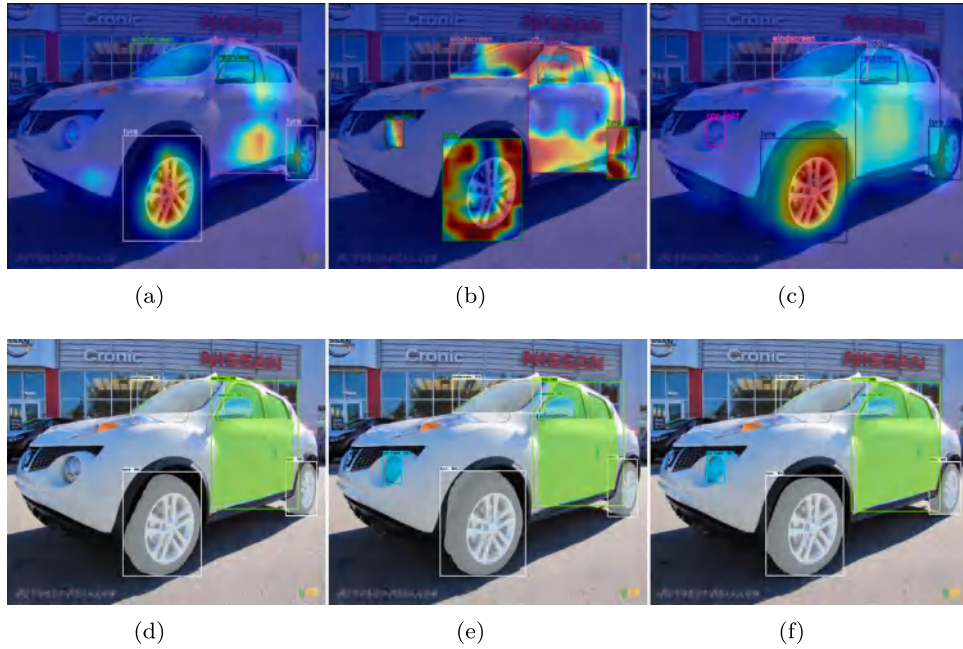


Fig. 13. (a), (b) and (c) are class activation diagrams of ResNet50+FPN, ResNet50+BHDF-FPN and ResNet50+BF-FPN, respectively. (d), (e) and (f) are the corresponding predictive segmentation diagrams.

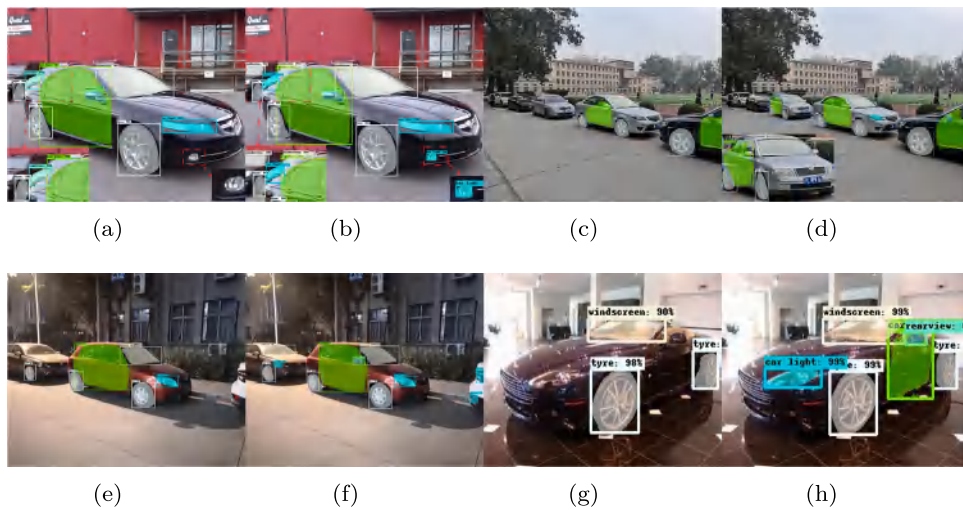


Fig. 14. This study evaluates the prediction performance of a model trained on VSD in complex scenarios. The detection results of the original model algorithm and the proposed algorithm are represented in (a), (c), (e), and (g) and (b), (d), (f), and (h), respectively.

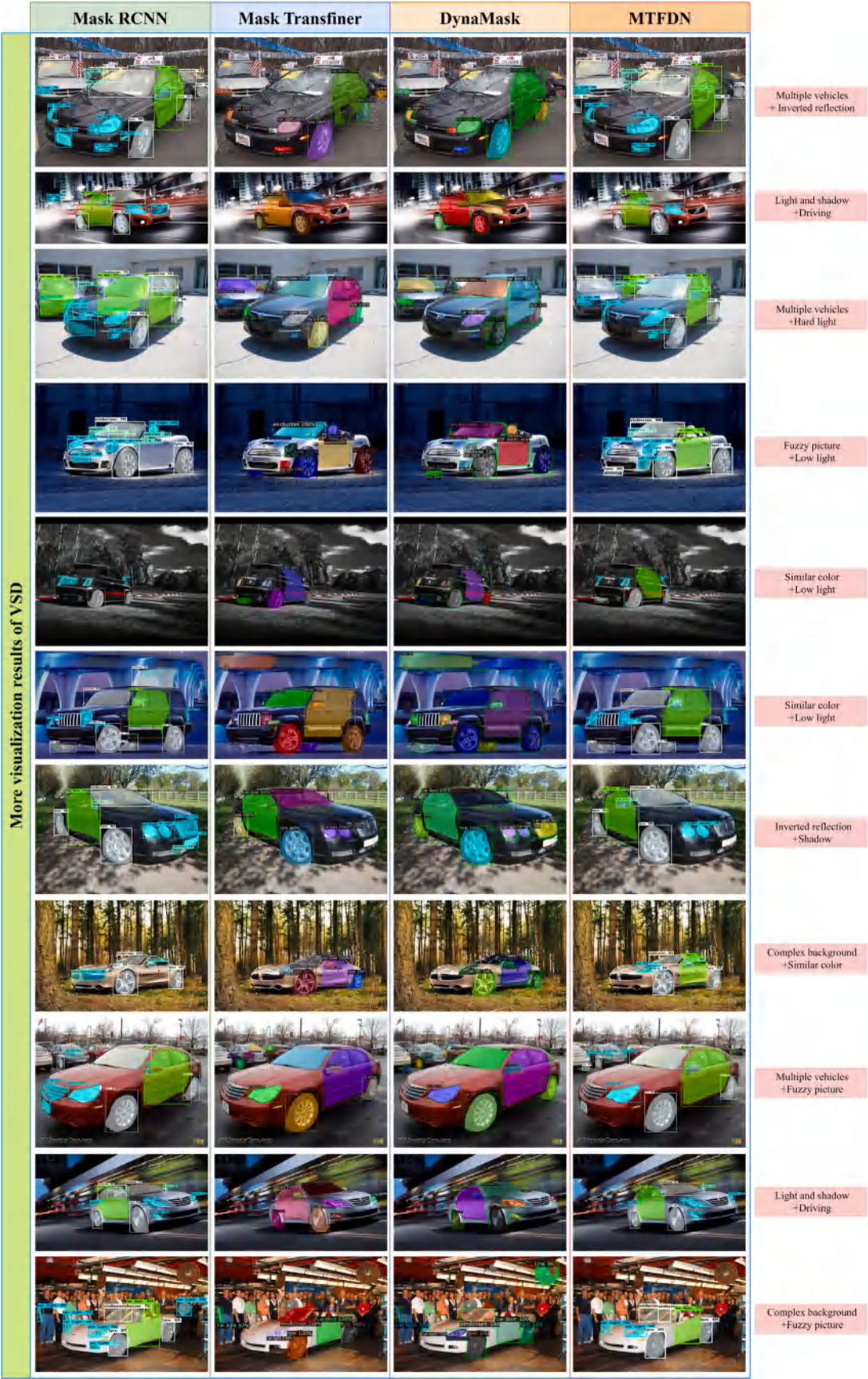


Fig. 15. More visualization results of VSD.



Fig. 16. Medium and long view of the vehicle component detection and segmentation effect based on VCD training model.

transformed by FPN into 5 tensors of different space sizes, which are stored as ordered dictionaries. To obtain the activation graph, we aggregate these image tensors, assign weights, and aggregate all parameters. We adjust tensors with different sizes into a common shape, connect them together, and highlight parts with high scores in coordinates and fractions to obtain the class activation diagram. Using the class activation diagram, we can understand, at the feature level, which parts of the input image were used to evaluate the class detection results.

An activation diagram comparison of each extraction network module is presented in Fig. 13.

In Fig. 13, More features are extracted in Fig. 13(b) and 13(c) than from Fig. 13(a), and headlight missing detection occurs in Fig. 13(d) compared with Fig. 13(e) and 13(f) in the prediction chart, which means that the BHDF-FPN and BF-FPN modules have stronger feature extraction and fusion ability than the FPN. Compared with Fig. 13(c), the features in Fig. 13(b) have more semantic information. For example, from the highlighted part of the class activation diagram, the tire is detected according to the tire's antiskid ring, and the door is detected according to the door's frame line, however, the features in Fig. 13(c) have more detailed information than those in Fig. 13(b) and the feature highlights in the class activation diagrams better fit the contours of the detected components, meanwhile, the prediction diagram Fig. 13(f) fits the component boundary better than the mask segmentation in Fig. 13(e). This also indicates that the BHDF-FPN is suitable for detection task in semantic feature extraction and fusion performance, while the BF-FPN avoids pixel aliasing caused by high-intensity fusion and is more suitable for pixel-level segmentation compared with the BHDF-FPN. This proves the rationality of using the BHDF-FPN for target detection and the BF-FPN for pixel segmentation in the single/dual network architecture design from the perspective of the class activation diagram feature visualization.

4.5. Effect display before and after improvement

4.5.1. Model training prediction effect based on VSD

The comparative prediction results are shown in Fig. 14.

Specifically, Fig. 14(a) and 14(b) represent the scenario where vehicles are blocking each other, and the proposed algorithm outperforms the original model algorithm in identifying the preset vehicle component category of the obstructed vehicles. Fig. 14(c) and 14(d) pertain to multivehicle scenarios, and the proposed algorithm demonstrates superior performance in identifying the preset vehicle component categories of multiple vehicles compared to the original model algorithm. Furthermore, Fig. 14(e) and 14(f) correspond to a dark scene scenario where the proposed algorithm outperforms the original model algorithm in identifying the missing components. Similarly, Fig. 14(g) and 14(h) represent situations where the picture quality is extremely fuzzy, and the proposed algorithm shows better performance in distinguishing the components with less obvious features in such scenarios compared to the original model algorithm.

Additionally, in daily life, various complex scenarios often occur simultaneously, posing a challenge to the performance requirements of algorithms. We have presented the prediction results of the MTFDN algorithm and some advanced algorithms in addressing multiple complex scenarios, as shown in Fig. 15. A comparison of the results indicates that the proposed algorithm demonstrates better detection and segmentation performance in recognizing these situations.

4.5.2. Medium and long view of the vehicle component detection and segmentation effect based on VCD training model

The comparative prediction results are shown in Fig. 16.

4.6. Application extension

In fact, it is worth noting that this decoupling of multi-task feature extraction and fusion is not restricted to Mask RCNN [42] but can be integrated into any top-down two-stage multi-task detection algorithm, such as Cascade Mask R-CNN [45], Hybrid Task Cascade(HTC [37]), and Mask Scoring R-CNN [44], to achieve superior performance compared to the baseline Mask RCNN. Cascade Mask R-CNN is a detection framework that is built upon the Mask R-CNN architecture and consists of a series of detectors with increasing IOU thresholds to gradually come true the prediction of the closer target. The detector is trained stage by stage, and the latter detector uses the output of the previous

Table 12

Based on VCD, the comparison experiment of MTFDN architecture applied to Cascade Mask R-CNN.

	Algorithm framework	MTFDN	$AP_{50}(\%)$		$AP_{50:95}(\%)$		$AR_{50:95}^{max\text{Dets}=100}(\%)$	
			Bbox	Segm	Bbox	Segm	Bbox	Segm
Single channel	Cascade Mask R-CNN	–	43.1	43.2	31.1	31.6	65.2	64.5
Dual channel		✓	49.3	48.8	35.6	35.3	69.3	68.7

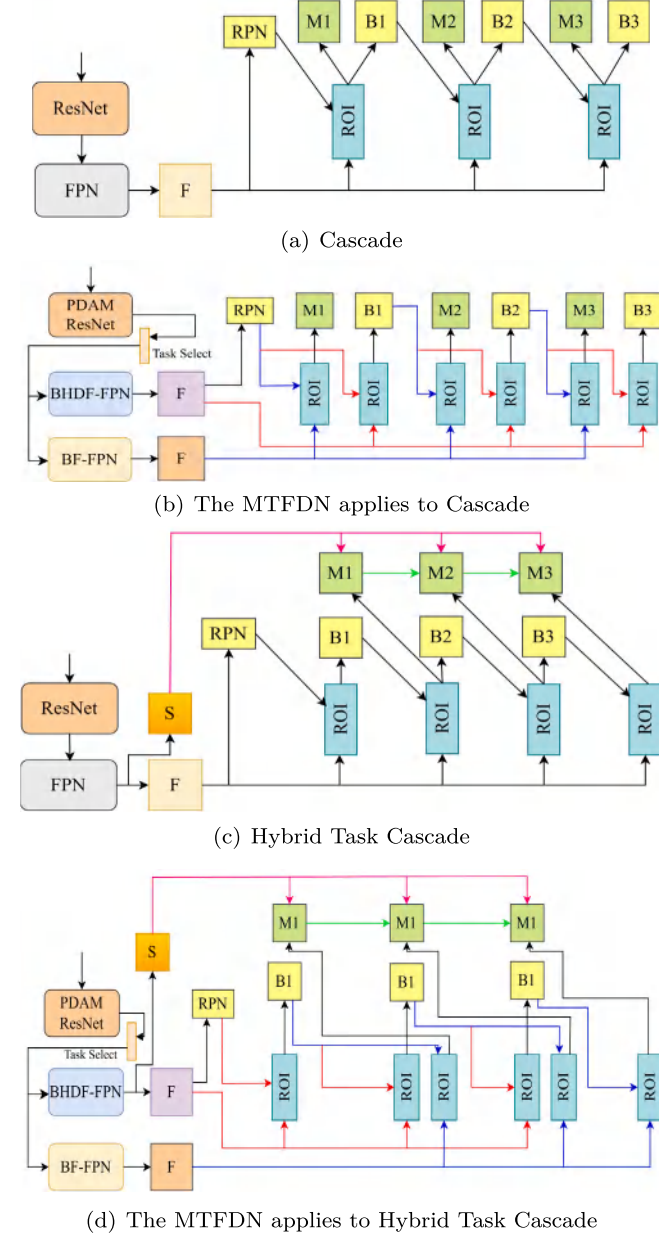


Fig. 17. The architecture of the different frameworks and the possible ways in which they apply the MTFDN. In the figure, “F” represents extracted features, “ROI” represents ROI Align or ROI pooling, “B” represents detection tasks, “M” represents segmentation tasks, the number represents Stage, and “S” represents the full convolutional network used for semantic segmentation tasks.

detector as the input, while Hybrid Task Cascade makes full use of the relationship between detection and segmentation on this basis. HTC further enhances the information flow based on Cascade Mask R-CNN and adds an additional feedforward semantic segmentation branch. This paper presents the network architecture diagram of MTFDN applied to Cascade Mask R-CNN and HTC, as shown in Fig. 17, and the algorithm applied to other top-down two-stage multi-task detection is also similar

In addition, a comparative experiment of Cascade Mask R-CNN is given in Table 12.

5. Conclusion

In response to the common problem of missed or false detection in complex scenarios caused by similar features and insufficient feature extraction in vehicle component detection, this study proposes for the first time an improved decoupled network architecture, the multi-task feature decoupling network (MTFDN), based on a multi-task top-down two-stage algorithm and designs three improved modules. Experimental results show that the MTFDN framework can effectively improve the aforementioned industry problems and outperforms the original model in terms of vehicle component detection and segmentation. Compared with various advanced detection and segmentation algorithms, the MTFDN framework still maintains advantages in model complexity and detection speed, and demonstrates even better performance in detection and segmentation.

In this study, we adopt a feature decoupling approach to optimize the most suitable feature extraction and fusion method according to the different detection and segmentation task characteristics, improving vehicle component detection and segmentation performance. This approach provides new ideas for multi-task coupling situations. At the same time, we reconstruct the attention trend of network learning features by imitating human cognitive processes through weight distribution, which makes feature learning more interpretable. Finally, we also give the expansion idea of applying MTFDN to Cascade Mask R-CNN and HTC, and the idea is the same in other top-down two-stage algorithms. We firmly believe that there is potential for further improvement of this method, and the better the feature extraction and fusion modules designed for detection and segmentation fit the requirements of a task, the better the decoupling effect of the features of the whole network.

Currently, the real-time performance of MTFDN is completely sufficient for practical application. The core point of contention in vehicle claims remains the accuracy of detection. The ability to accurately identify and segment various vehicle components in images determines the reliability and social value of the intelligent claims assessment system. If there is a future need for deployment on mobile devices, the requirements for model size and algorithm complexity must also be considered. Therefore, we will continue our research to further improve the accuracy of component detection and segmentation networks and optimize algorithm complexity. In addition, we will explore the integration of vehicle component detection and damage detection in order to achieve better deployment and utilization.

CRedit authorship contribution statement

Yongjie Zhai: Supervision, Project administration, Conceptualization. **Xunqi Zhou:** Writing – original draft, Software, Methodology. **Nianhao Chen:** Visualization, Validation. **Xuan Liu:** Writing – review & editing, Investigation. **Zhenqi Zhang:** Writing – review & editing, Investigation. **Xinying Wang:** Formal analysis, Data curation. **Qianming Wang:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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